



National Institute of Technology Rourkela
Department of Computer Science Engineering
Deep Learning (CS6510)
End-Semester Examination
B.Tech 4th year/PhD/M.Tech(R)

Duration: 3 hrs

Spring 2026, 8th Semester

Total marks: 50

Attempt all questions;

⟨CO-x: Course Outcome level⟩; ⟨BL-x: Bloom's taxonomy level⟩;

**All parts of the same question must be answered together*

1. (a) Given filter of width F and depth d , applied in an image with dimension $(W1, H1)$. Derive the equation for finding $(W2, H2)$ assuming that there are S strides and P padding during the first convolution operation. If there are K kernels in the ANN, what will be the number of parameters formed at the same time. ⟨CO-4,5⟩ ⟨BL-2, 3⟩ [5]
- (b) Using a single-layer perceptron, simulate the following combinatorial circuit elements. Find the Weight vectors (W_i) and Bias (β) for each of the following. The circuit has two inputs x_1, x_2 . ⟨CO-1,2⟩ ⟨BL-2, 3⟩ [5]
 - i) OR gate
 - ii) AND gate
 - iii) OR gate followed by an AND gate with the second gate (OR) receiving two inputs: first as the output from the AND gate, and second from the source input x_2 .
2. (a) Demonstrate the complete LeNet-5 architecture for handwritten character recognition, depicting all the variable values involved in every layer transition. Parameters: Stride (S), Kernel width (F), Kernels (K), Padding (P), Parameters ($Param$) ⟨CO-4⟩ ⟨BL-3⟩ [5]
- (b) Using back-propagation for the neural network in Fig. 1, compute the output of nodes y_3, y_4, y_5 . Assume the activation function to be a sigmoid. Data: #Epochs = 2; Target output = 0.35; Learning rate = 0.01; Loss function = MSE; ⟨CO-4,5⟩ ⟨BL-5⟩ [5]

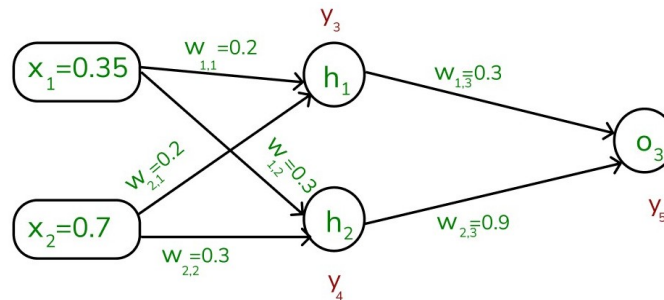


Figure 1: Neural network

3. (a) Explain the bias-variance trade-off with an appropriate diagram? Derive the equation for predicted error in terms of bias and variance. ⟨CO-3⟩ ⟨BL-3⟩ [5]
 (b) Assertion: Given an input image I of size $n \times n$, a convolution operation is performed upon I with a filter of dimension $k \times k$ twice. Once with $k = 1$ and then with $k = n$. What will be the “Conclusion(s)” under each of these cases? Also mention the “Reason(s)” as to why you draw the given conclusions? ⟨CO-3⟩ ⟨BL-3⟩ [5]
4. (a) Derive the generic term to calculate $\text{grad}(w_i)$ from the given ANN below (Fig. 2). Subsequently find the expressions for $\text{grad}(w_1)$, $\text{grad}(w_2)$, $\text{grad}(w_3)$. ⟨CO-3⟩ ⟨BL-3⟩ [5]

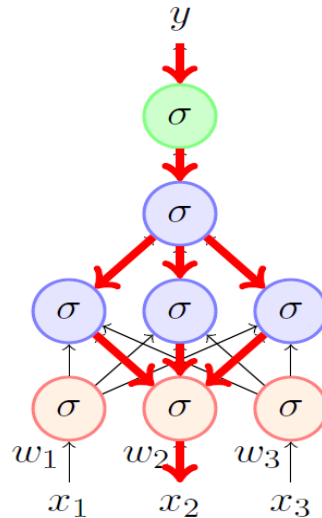


Figure 2: Sample ANN

- (b) Assertion: Given an DNN (Deep NN) of multiple layers. Suppose the core computation happens in only 10% of the layers while 90% of the layers are deemed as redundant. ⟨CO-3,4⟩ ⟨BL-4,5⟩ [5]

Conclusion(s) (May be True/False): The ANN can be made efficient by using only the 10% of the active layers. The backpropagation will not have a major impact on the remaining layers.

Reason(s): —

5. (a) Compare and contrast the following CNN architectures: LeNet5, AlexNet, ResNet, ZFNet, VGGNet for the MNIST image dataset. ⟨CO-5⟩ ⟨BL-4⟩ [5]
 (b) Given an input image of dimensions $227 \times 227 \times 3$; For Convo1 layer: $K = 96, F = 11, S = 4, P = 0$. Find W_2, H_2 and the number of parameters involved. ⟨CO-5⟩ ⟨BL-5⟩ [5]

N.B: Make your own logical assumptions (if required) to answer wherever necessary.